1 The Comparison Test

Given an improper integral $\int_a^\infty f(x)dx$, we saw that we can determine whether or not it converges by comparing to an improper integral $\int_a^\infty g(x)dx$ which is known to converge or diverge. Here, we follow the same strategy to determine whether a given series $\sum a_n$ converges or diverges.

Theorem 1.1 (The Comparison Test)**.** *Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms.*

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum a_n$ is also divergent.

Before proving the Comparison Test, we will get some practice using it. In using the Comparison Test, of course, have some known series $\sum b_n$ for the purpose of comparison. Most of the time we use one of these series:

- A p-series (Recall that $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.)
- A geometric series (Recall that $\sum ar^{n-1}$ converges if $|r| < 1$ and diverges if $|r| \geq 1$.)

1. Use the Comparison Test to prove that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n+1}$ converges.

2. Determine whether the series $\displaystyle\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ converges or diverges.

The condition $0 \leq a_n \leq b_n$ (or $0 \leq b_n \leq a_n$) in the Comparison Test really only needs to hold for all $n \geq N$, where N is some fixed positive integer, because the convergence of a series is not affected by finitely many terms. This is illustrated in the next example.

3. Determine whether $\displaystyle\sum_{k=1}^{\infty} \frac{\ln(k)}{k}$ converges or diverges.

We will now prove the Comparison Test.

Proof. Let $s_n = \sum_{i=1}^{n} a_i$ and $t_n = \sum_{i=1}^{n} b_i$.

4. Show that the sequences $\{s_n\}$ and $\{t_n\}$ are increasing.

5. Part (i): Suppose $a_i \leq b_i$ for all $i \in \mathbb{N}$ and that $\sum b_i$ converges. Let $t = \sum_{i=1}^{\infty} b_n$. Prove $\{s_n\}$ is bounded above by t.

6. Explain why your answers to questions 4 and 5 imply that $\sum a_n$ converges.

7. Part (ii): Suppose $b_i \leq a_i$ for all $i \in \mathbb{N}$ and that $\sum b_i$ diverges. Since t_n is increasing, we must have $t_n \to \infty$ as $n \to \infty$. Show that this implies $\sum a_i$ diverges.

□

2 The Limit Comparison Test

If the terms of a series are larger than those of a convergent series or smaller than those of a divergent series, then the Comparison Test does not apply. Consider, for instance, the series

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1}.$$

The inequality

$$\frac{1}{2^n - 1} > \frac{1}{2^n}$$

is useless as far as the Comparison Test is concerned because $\sum_n b_n = \sum_n \left(\frac{1}{2}\right)^n$ is convergent and $a_n > b_n$. Nonetheless, $\sum_{n=1}^{\infty} \frac{1}{2^n-1}$ ought to be convergent, because it is very similar to the convergent geometric series $\sum \left(\frac{1}{2}\right)^n$. As we did for improper integrals, in such cases we can use the *Limit Comparison Test*.

Theorem 2.1 (Limit Comparison Test)**.** *Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms. If*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c,$$

where c is a positive real number, then either both series converge or both series diverge.

Proof. Let $0 < \varepsilon < c$, so that $c - \varepsilon > 0$. Then, since $\lim_{n\to\infty} \frac{a_n}{b_n} = c$, corresponding to ε there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|\frac{a_n}{b_n} - c| < \varepsilon$, or, equivalently, $0 < c - \varepsilon < \frac{a_n}{b_n} < c + \varepsilon$. Multiplying both sides by b_n, we have

$$0 < (c - \varepsilon)b_n < a_n < (c + \varepsilon)b_n. \tag{2.2}$$

8. Use the Comparison Test to finish the proof.

□

9. Use the Limit Comparison Test to show that the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ indeed converges.

10. Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$ converges or diverges.

We can extend the Limit Comparison Test as follows when $c = 0$ or $c = \infty$.

Theorem 2.3 (Extension of Limit Comparison Test)**.** *Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms.*

(i) If $\sum b_n$ is convergent and $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$, then $\sum a_n$ is also divergent.

11. Prove Theorem 2.3.

12. Determine whether each of the following series converges or diverges.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln(n)}{n^3}$

(b) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln(n)}{\sqrt{n}e^n}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$